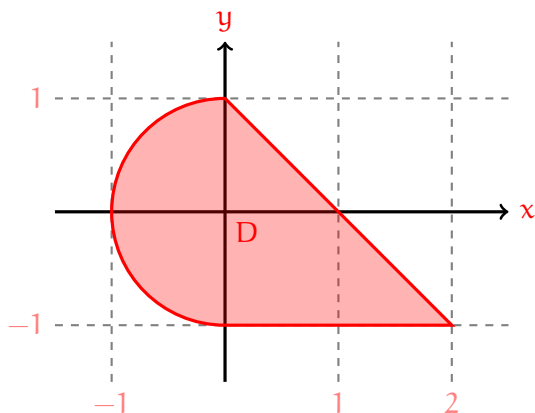


Example 1. Let D be the region below, consisting of a half-disk and triangle.



(a) Set up $\iint_D f(x, y) \, dA$ using an iterated integral in order $dx dy$.

Solution. In order $dx dy$ we first calculate xz cross-sectional areas of the volume.

For any fixed y satisfying:

we have:

Thus

$$\iint_D f(x, y) \, dA =$$

(b) Set up $\iint_D f(x, y) \, dA$ using an iterated integral in order $dy dx$.

Solution. In order $dy dx$ we first calculate yz cross-sectional areas of the volume.

For any fixed x satisfying:

we have:

Whereas for any fixed x satisfying:

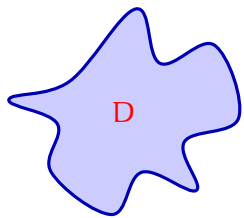
we have:

Thus

$$\iint_D f(x, y) \, dA =$$

A. **Area and Density.** Let's consider another way to view double integrals.

Suppose D represents a **lamina** (a thin plate) in the xy -plane.



Assume its **density** at point (x, y) is given by

Then the mass of an infinitesimal bit of area dA is:

We add up these infinitesimal masses to obtain the total mass.

[total mass of D] =

More succinctly

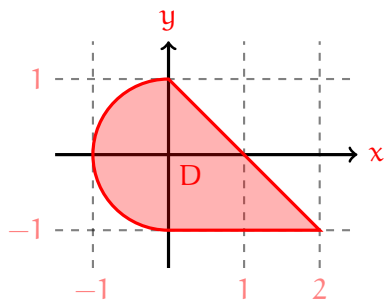
$$\iint_D [\text{density}] \, dA =$$

As a special case, imagine that $[\text{density}] = 1$, or in other words:

Then

$$\iint_D 1 \, dA =$$

Example 2. Find $\iint_D 1 \, dA$ for the region D sketched below.



Solution. The double-integral of 1 equals

And thus

$$\iint_D 1 \, dA =$$

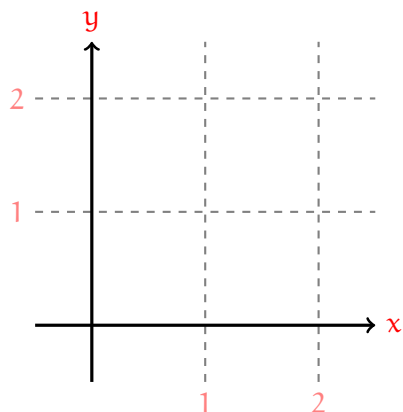
Example 3. Find $\int_0^2 \int_y^2 e^{x^2} dx dy$.

Solution. In the given order we would need

$$\int e^{x^2} dx =$$

Let's instead change order of integration to $dy dx$.

The region of integration is determined by bounds



For order $dy dx$ we first calculate yz cross-sectional areas of volume.

For any fixed x satisfying:

we have:

Thus

$$\int_0^2 \int_y^2 e^{x^2} dx dy =$$

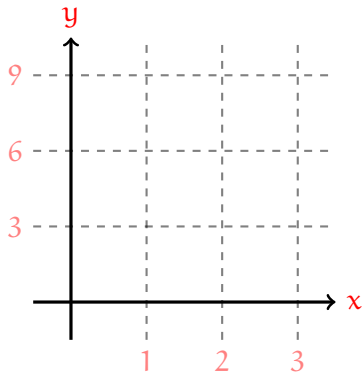
Example 4. Find the integral by changing order.

$$\int_0^3 \int_0^{9-x^2} \frac{xe^{3y}}{9-y} dy dx$$

Solution. The given order would be impossible using elementary functions.

As commanded, let's instead change order of integration to $dx dy$.

The region of integration is determined by bounds



For order $dx dy$ we first calculate xz cross-sectional areas.

For any fixed y satisfying:

we have:

Thus

$$\int_0^3 \int_0^{9-x^2} \frac{xe^{3y}}{9-y} dy dx =$$